Type MS · Caesar cipher

Answer 1 of 4 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part a.i in this mark scheme extract

4(a)(i)	A1 A2	Award **one** mark for each correct cell: **Plaintext** / **Shift** / **Ciphertext** PIXEL / -4 / LETAH CLOUD / +3 / FORXG	Allow lower case ciphertext	**2**

Question	mp	Answer	Additional Guidance	Mark
4(a)(ii)	A3 A4	Award **two** marks for a comparison such as: • Both produce same result (IQNF)(1) because the shift of -6 followed by +8 is the same as the shift of +2 (1) • The result is the same (IQNF) (1), but the single shift is more efficient than a double shift (1) Practical explanation: GOLD (-6) = AIFX (+8) = IQNF (1) GOLD (+2) = IQNF (1)	can have 1 mark from MP and 1 mark from Practical e.g. GOLD (+2) = IQNF (1) which would be same result (1)	**2**
4(a)(iii)	A5 A6	Award **two** marks for a linked explanation such as: The letter Y has been encrypted incorrectly as the letter V / the letter Y should have been encrypted to the letter D (1), because the +5 shift did not roll over the end of the alphabet correctly (1)	identify where error occurred (1) and can explain how error occurred (1)	**2**

Type MS · Caesar cipher

Answer 1 of 4 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
4(b)	Award **one** mark for each of:		B1 does not require casting for data types on input B6 do **not** accept taking just first digit from a string B8 is only for print(), key may not be correct	8
	B1	Word and 2 numbers input		
	B2	Selection/loop used to check word input		
	B3	Test for word length exactly 2		
	B4	Appropriate error message for invalid string input		
	B5	Reverse the inputted word, any method		
	B6	Whole number part of the decimal number generated		
	B7	Correct key generated (concatenation)		
	B8	Key output		

Suggested test data and output			
word	**int**	**float**	**Output**
qwe/q			error message
qw	12	17.89	12wq17

Type MS · Caesar cipher

Answer 1 of 4 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part a.i in this mark scheme extract

C#

```
// Create the variables
String myWord = "";
String newWord = "";
int myNum = 0;
double myDecimal = 0.0;
String myKey = "";
int myWhole = 0;

// Take a word as input
Console.Write("Enter a word: ");
myWord = Console.ReadLine();

// Check that the word is correct length
if (myWord.Length == 2)
{
    // Take a whole number as input
    Console.Write("Enter a whole number: ");
    myNum= Convert.ToInt32(Console.ReadLine());

    // Take a decimal number as input
    Console.Write("Enter a decimal number: ");
    myDecimal = Convert.ToDouble(Console.ReadLine());

    // Reverse the letters in the word
    newWord = myWord[1].ToString() + myWord[0].ToString();

    // Find the whole number part of the decimal number
    myWhole = (int)myDecimal;

    // Create the new key
    myKey = String.Concat(myNum, newWord, myWhole);

    // Display the new key for the user
    Console.Write("The key is " + myKey);
}
else
{
    // Word is not the correct length, so display an error message
    Console.WriteLine("Invalid word entered");
}
```

Type MS · Caesar cipher

Answer 1 of 4 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part a.i in this mark scheme extract

Java

```
// Write your code below this line

Scanner myScanner = new Scanner (System.in);

// Create the variables
String myWord = "";
String newWord = "";
int myNum = 0;
Double myDecimal = 0.0;
String myKey = "";
int myWhole = 0;

// Take a word as input
System.out.print("Enter a word: ");
myWord = myScanner.nextLine();

// Check that the word is correct length
if (myWord.length() == 2)
{
    // Take a whole number as input
    System.out.print("Enter a whole number: ");
    myNum = Integer.parseInt(myScanner.nextLine());

    // Take a decimal number as input
    System.out.print("Enter a decimal number: ");
    myDecimal = Double.parseDouble (myScanner.nextLine());

    // Reverse the letters in the word
    newWord = Character.toString(myWord.charAt(1)) +
              Character.toString(myWord.charAt(0));

    // Find the whole number part of the decimal number
    myWhole = myDecimal.intValue();

    // Create the new key
    myKey = myNum + newWord + myWhole;

    // Display the new key for the user
    System.out.println("The key is " + myKey);
}
else
{
    // Word is not the correct length, so display an error message
    System.out.println("Invalid word entered");
}
}
```

Type MS · Caesar cipher

Answer 1 of 4 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part a.i in this mark scheme extract

Python

```
# Take a two-letter word as input
myWord = input ("Enter a word: ")

# Check that the word is correct length
if (len (myWord) == 2):

    # Take a whole number as input
    myNum = int (input ("Enter a whole number: "))

    # Take a decimal number as input
    myDecimal = float (input ("Enter a decimal number: "))

    # Reverse the letters in the word
    newWord = myWord[1] + myWord[0]

    # Find the whole number part of the decimal number
    myWhole = int (myDecimal)

    # Create the new key
    myKey = str (myNum) + newWord + str (myWhole)

    # Display the new key for the user
    print (myKey)

# Word is not the correct length, so display an error message
else:
    print ("Invalid word entered")
```

Type MS · Caesar cipher

Answer 1 of 4 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 6 of 6 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 1 of 8 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (b)	B1	Award up to 2 marks for a linked explanation such as: • The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1) • It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1)	Accept alternative similar wording.	(2)

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 2 of 8 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (c)		Award 1 mark each up to a maximum of 4 for: Encrypted letter: f l m k t r w h e e Keyword letter: t h i r t y t h i r Decrypted letter: m e e t a t d a w n		
	C1	Award 1 mark from: • flmktrwhee mapped to thirtythir (1) • Flmktr mapped to thirty (1)		
	C2	At least one letter decrypted correctly (1)		
	C3	At least one word decrypted correctly (1)		
	C4	Decrypted message 'meet at dawn' (1)		(4)

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 3 of 8 · find part b in this mark scheme extract

Question	IMP	Answer	Additional Guidance	Mark
3 (d)(i)	D1	Award 1 mark for: • cipher[letter / a single encrypted letter (1)	Do not accept word/message/text	(1)

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 4 of 8 · find part b in this mark scheme extract

		• cipherLetter / a single encrypted letter (1)		(1)
3 (d)(ii)	D2	Award 1 mark for any of: • keywordLetter • plaintextLetter	Ignore case	(1)

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 5 of 8 · find part b in this mark scheme extract

		• plaintextLetter		(1)
3 (d)(iii)	D2	Award 1 mark for any of: • subprogram that is already defined • subprogram that is already written • subprogram that is already compiled • subprogram that can be called without having to write code for it	Must be clear they are referring to subprogram provided by the language itself	(1)

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 6 of 8 · find part b in this mark scheme extract

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 7 of 8 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (a)	A1	Get plaintext and store in plaintext variable	Accept alternative wording Line numbers may be different compared to the examples shown When testing the completed code use lowercase for the input	(1)
	A2	Get key and store in key variable		(1)
	A3	Validate key (must repeat until a valid key has been input)		(1)
	A4	Open file to write/append		(1)
	A5	Write cipher text		(1)
	A6	Close file		(1)
	A7	Displays ciphertext		(1)
	A8	Executing and producing correct output to file and screen (must have A3)		(1)
Code examples				
Python				

```
#       Add your code to get the plaintext and convert it to lowercase
plaintext = input("Enter the plaintext using lowercase letters: ")
plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
# Add your code to write the ciphertext to a text file
cipherFile = open("Cipher.txt", "w")
cipherFile.write(ciphertext)
cipherFile.close()
```

```
#       Add your code to display the ciphertext
print("The ciphertext message is:",ciphertext)
```

Type MS · Caesar cipher

Answer 2 of 4 · Question 3 part b · 2 marks

2021 JUN · Paper 02 · extract 8 of 8 · find part b in this mark scheme extract

Type MS · Caesar cipher

Answer 3 of 4 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part a.ii in this mark scheme extract

4(a)(i)	A1 A2	Award **one** mark for each correct cell: **Plaintext** / **Shift** / **Ciphertext** PIXEL / -4 / LETAH CLOUD / +3 / FORXG	Allow lower case ciphertext	**2**

Question	mp	Answer	Additional Guidance	Mark
4(a)(ii)	A3 A4	Award **two** marks for a comparison such as: • Both produce same result (IQNF)(1) because the shift of -6 followed by +8 is the same as the shift of +2 (1) • The result is the same (IQNF) (1), but the single shift is more efficient than a double shift (1) Practical explanation: GOLD (-6) = AIFX (+8) = IQNF (1) GOLD (+2) = IQNF (1)	can have 1 mark from MP and 1 mark from Practical e.g. GOLD (+2) = IQNF (1) which would be same result (1)	**2**
4(a)(iii)	A5 A6	Award **two** marks for a linked explanation such as: The letter Y has been encrypted incorrectly as the letter V / the letter Y should have been encrypted to the letter D (1), because the +5 shift did not roll over the end of the alphabet correctly (1)	identify where error occurred (1) and can explain how error occurred (1)	**2**

Type MS · Caesar cipher

Answer 3 of 4 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
4(b)	Award **one** mark for each of:		B1 does not require casting for data types on input B6 do **not** accept taking just first digit from a string B8 is only for print(), key may not be correct	
	B1	Word and 2 numbers input		
	B2	Selection/loop used to check word input		
	B3	Test for word length exactly 2		
	B4	Appropriate error message for invalid string input		
	B5	Reverse the inputted word, any method		
	B6	Whole number part of the decimal number generated		
	B7	Correct key generated (concatenation)		
	B8	Key output		**8**

Suggested test data and output			
word	**int**	**float**	**Output**
qwe/q			error message
qw	12	17.89	12wq17

Type MS · Caesar cipher

Answer 3 of 4 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part a.ii in this mark scheme extract

C#

```
// Create the variables
String myWord = "";
String newWord = "";
int myNum = 0;
double myDecimal = 0.0;
String myKey = "";
int myWhole = 0;

// Take a word as input
Console.Write("Enter a word: ");
myWord = Console.ReadLine();

// Check that the word is correct length
if (myWord.Length == 2)
{
    // Take a whole number as input
    Console.Write("Enter a whole number: ");
    myNum= Convert.ToInt32(Console.ReadLine());

    // Take a decimal number as input
    Console.Write("Enter a decimal number: ");
    myDecimal = Convert.ToDouble(Console.ReadLine());

    // Reverse the letters in the word
    newWord = myWord[1].ToString() + myWord[0].ToString();

    // Find the whole number part of the decimal number
    myWhole = (int)myDecimal;

    // Create the new key
    myKey = String.Concat(myNum, newWord, myWhole);

    // Display the new key for the user
    Console.Write("The key is " + myKey);
}
else
{
    // Word is not the correct length, so display an error message
    Console.WriteLine("Invalid word entered");
}
```

Type MS · Caesar cipher

Answer 3 of 4 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part a.ii in this mark scheme extract

Java

```
    // Write your code below this line

    Scanner myScanner = new Scanner (System.in);

    // Create the variables
    String myWord = "";
    String newWord = "";
    int myNum = 0;
    Double myDecimal = 0.0;
    String myKey = "";
    int myWhole = 0;

    // Take a word as input
    System.out.print("Enter a word: ");
    myWord = myScanner.nextLine();

    // Check that the word is correct length
    if (myWord.length() == 2)
    {
        // Take a whole number as input
        System.out.print("Enter a whole number: ");
        myNum = Integer.parseInt(myScanner.nextLine());

        // Take a decimal number as input
        System.out.print("Enter a decimal number: ");
        myDecimal = Double.parseDouble (myScanner.nextLine());

        // Reverse the letters in the word
        newWord = Character.toString(myWord.charAt(1)) +
                  Character.toString(myWord.charAt(0));

        // Find the whole number part of the decimal number
        myWhole = myDecimal.intValue();

        // Create the new key
        myKey = myNum + newWord + myWhole;

        // Display the new key for the user
        System.out.println("The key is " + myKey);
    }
    else
    {
        // Word is not the correct length, so display an error message
        System.out.println("Invalid word entered");
    }
}
```

Type MS · Caesar cipher

Answer 3 of 4 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part a.ii in this mark scheme extract

Python

```
# Take a two-letter word as input
myWord = input ("Enter a word: ")

# Check that the word is correct length
if (len (myWord) == 2):

    # Take a whole number as input
    myNum = int (input ("Enter a whole number: "))

    # Take a decimal number as input
    myDecimal = float (input ("Enter a decimal number: "))

    # Reverse the letters in the word
    newWord = myWord[1] + myWord[0]

    # Find the whole number part of the decimal number
    myWhole = int (myDecimal)

    # Create the new key
    myKey = str (myNum) + newWord + str (myWhole)

    # Display the new key for the user
    print (myKey)

# Word is not the correct length, so display an error message
else:
    print ("Invalid word entered")
```

Type MS · Caesar cipher

Answer 3 of 4 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 6 of 6 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Caesar cipher

Answer 4 of 4 · Question 4 part a.iii · 2 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part a.iii in this mark scheme extract

<table>
<tr><td>4(a)(i)</td><td>A1
A2</td><td>Award one mark for each correct cell:

<table><tr><th>Plaintext</th><th>Shift</th><th>Ciphertext</th></tr><tr><td>PIXEL</td><td>-4</td><td>LETAH</td></tr><tr><td>CLOUD</td><td>+3</td><td>FORXG</td></tr></table></td><td>Allow lower case ciphertext</td><td>2</td></tr>
</table>

Question	mp	Answer	Additional Guidance	Mark
4(a)(ii)	A3 A4	Award **two** marks for a comparison such as: • Both produce same result (IQNF)(1) because the shift of -6 followed by +8 is the same as the shift of +2 (1) • The result is the same (IQNF) (1), but the single shift is more efficient than a double shift (1) Practical explanation: GOLD (-6) = AIFX (+8) = IQNF (1) GOLD (+2) = IQNF (1)	can have 1 mark from MP and 1 mark from Practical e.g. GOLD (+2) = IQNF (1) which would be same result (1)	**2**
4(a)(iii)	A5 A6	Award **two** marks for a linked explanation such as: The letter Y has been encrypted incorrectly as the letter V / the letter Y should have been encrypted to the letter D (1), because the +5 shift did not roll over the end of the alphabet correctly (1)	identify where error occurred (1) and can explain how error occurred (1)	**2**

Type MS · Caesar cipher

Answer 4 of 4 · Question 4 part a.iii · 2 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part a.iii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
4(b)	Award **one** mark for each of:		B1 does not require casting for data types on input B6 do **not** accept taking just first digit from a string B8 is only for print(), key may not be correct	8
	B1	Word and 2 numbers input		
	B2	Selection/loop used to check word input		
	B3	Test for word length exactly 2		
	B4	Appropriate error message for invalid string input		
	B5	Reverse the inputted word, any method		
	B6	Whole number part of the decimal number generated		
	B7	Correct key generated (concatenation)		
	B8	Key output		

Suggested test data and output			
word	**int**	**float**	**Output**
qwe/q			error message
qw	12	17.89	12wq17

Type MS · Caesar cipher

Answer 4 of 4 · Question 4 part a.iii · 2 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part a.iii in this mark scheme extract

C#

```
// Create the variables
String myWord = "";
String newWord = "";
int myNum = 0;
double myDecimal = 0.0;
String myKey = "";
int myWhole = 0;

// Take a word as input
Console.Write("Enter a word: ");
myWord = Console.ReadLine();

// Check that the word is correct length
if (myWord.Length == 2)
{
    // Take a whole number as input
    Console.Write("Enter a whole number: ");
    myNum= Convert.ToInt32(Console.ReadLine());

    // Take a decimal number as input
    Console.Write("Enter a decimal number: ");
    myDecimal = Convert.ToDouble(Console.ReadLine());

    // Reverse the letters in the word
    newWord = myWord[1].ToString() + myWord[0].ToString();

    // Find the whole number part of the decimal number
    myWhole = (int)myDecimal;

    // Create the new key
    myKey = String.Concat(myNum, newWord, myWhole);

    // Display the new key for the user
    Console.Write("The key is " + myKey);
}
else
{
    // Word is not the correct length, so display an error message
    Console.WriteLine("Invalid word entered");
}
```

Type MS · Caesar cipher

Answer 4 of 4 · Question 4 part a.iii · 2 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part a.iii in this mark scheme extract

Java

```
// Write your code below this line

Scanner myScanner = new Scanner (System.in);

// Create the variables
String myWord = "";
String newWord = "";
int myNum = 0;
Double myDecimal = 0.0;
String myKey = "";
int myWhole = 0;

// Take a word as input
System.out.print("Enter a word: ");
myWord = myScanner.nextLine();

// Check that the word is correct length
if (myWord.length() == 2)
{
    // Take a whole number as input
    System.out.print("Enter a whole number: ");
    myNum = Integer.parseInt(myScanner.nextLine());

    // Take a decimal number as input
    System.out.print("Enter a decimal number: ");
    myDecimal = Double.parseDouble (myScanner.nextLine());

    // Reverse the letters in the word
    newWord = Character.toString(myWord.charAt(1)) +
              Character.toString(myWord.charAt(0));

    // Find the whole number part of the decimal number
    myWhole = myDecimal.intValue();

    // Create the new key
    myKey = myNum + newWord + myWhole;

    // Display the new key for the user
    System.out.println("The key is " + myKey);
}
else
{
    // Word is not the correct length, so display an error message
    System.out.println("Invalid word entered");
}
}
```

Type MS · Caesar cipher

Answer 4 of 4 · Question 4 part a.iii · 2 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part a.iii in this mark scheme extract

Python

```
# Take a two-letter word as input
myWord = input ("Enter a word: ")

# Check that the word is correct length
if (len (myWord) == 2):

    # Take a whole number as input
    myNum = int (input ("Enter a whole number: "))

    # Take a decimal number as input
    myDecimal = float (input ("Enter a decimal number: "))

    # Reverse the letters in the word
    newWord = myWord[1] + myWord[0]

    # Find the whole number part of the decimal number
    myWhole = int (myDecimal)

    # Create the new key
    myKey = str (myNum) + newWord + str (myWhole)

    # Display the new key for the user
    print (myKey)

# Word is not the correct length, so display an error message
else:
    print ("Invalid word entered")
```

Type MS · Caesar cipher

Answer 4 of 4 · Question 4 part a.iii · 2 marks

2023 JUN · Paper 02 · extract 6 of 6 · find part a.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark